

Machine Learning-Based Late Delivery Risk Prediction in Global Supply Chain Operations

Research Paper

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Abstract

Late deliveries can cause service-level breaches, customer dissatisfaction, additional logistics costs, and downstream disruption. This study develops a machine-learning framework for predicting late-delivery risk using an APL Logistics order dataset containing 180,519 observations and 40 original variables. The workflow combines exploratory data analysis, business-oriented feature engineering, preprocessing, supervised classification, model comparison, and explainable artificial intelligence. Logistic Regression, Random Forest, and XGBoost were trained using an 80:20 stratified train-test split. XGBoost achieved the best ROC-AUC of 0.7601, followed by Logistic Regression at 0.7552 and Random Forest at 0.7536. On the held-out test set, XGBoost achieved 0.8449 precision, 0.5574 recall, 0.6717 F1-score, and 70.13% accuracy. SHAP analysis was used to examine feature contributions, and a Streamlit dashboard translates predictions into practical risk categories. The study demonstrates how predictive analytics can support proactive logistics intervention and supply-chain decision making.

Keywords: Supply Chain Analytics; Late Delivery Prediction; Machine Learning; XGBoost; SHAP; Logistics Risk; Predictive Analytics; Streamlit

1. Introduction

Global supply-chain operations involve interacting decisions related to transportation, order processing, inventory, customer demand, geography, and service levels. A late shipment can create consequences beyond one order. At large order volumes, manually identifying every potentially late order is difficult and reactive.

Predictive analytics enables a shift from reactive exception handling toward proactive risk management. A classification model can estimate the probability that an order belongs to a late-delivery-risk class, supporting prioritization, customer communication, routing review, and escalation.

This research develops an end-to-end machine-learning solution that combines predictive performance with explainability and an interactive dashboard.

2. Problem Statement

The central problem is the early identification of orders likely to experience late delivery. The model should use information available for prediction while avoiding variables that directly reveal the final delivery outcome.

The solution should provide useful discrimination, identify influential factors, and present predictions in a form that can support logistics decision making.

3. Objectives

- Analyze late-delivery risk across important logistics and customer dimensions.
- Create business-oriented features representing shipping pressure, quantity, discount, express shipping, order complexity, and profitability.
- Compare Logistic Regression, Random Forest, and XGBoost.
- Evaluate models using ROC-AUC, precision, recall, F1-score, accuracy, and a confusion matrix.
- Use SHAP-based explainability to examine feature contributions.
- Develop a Streamlit decision-support interface with Low, Medium, and High risk categories.

4. Dataset and Data Preparation

The dataset contains 180,519 records and 40 original columns covering operational, commercial, customer, product, market, geographic, and shipping information. The target is **Late_delivery_risk**, where 1 represents late-delivery risk and 0 represents the non-risk class.

The target contains 98,977 positive observations and 81,542 negative observations, approximately 54.8% and 45.2%. The class distribution is therefore not severely imbalanced.

The CSV was read using Latin-1 encoding. Numerical variables were processed with median imputation and scaling, while categorical variables used most-frequent imputation and one-hot encoding with unknown categories ignored.

4.1 Data Leakage Prevention

The variables **Days for shipping (real)** and **Delivery Status** were excluded because they reveal post-outcome information. Removing them prevents leakage and makes the predictive setup more realistic.

5. Exploratory Data Analysis

EDA examined the target distribution and differences by shipping mode, order region, customer segment, and scheduled shipping days. The following are the actual project figures.

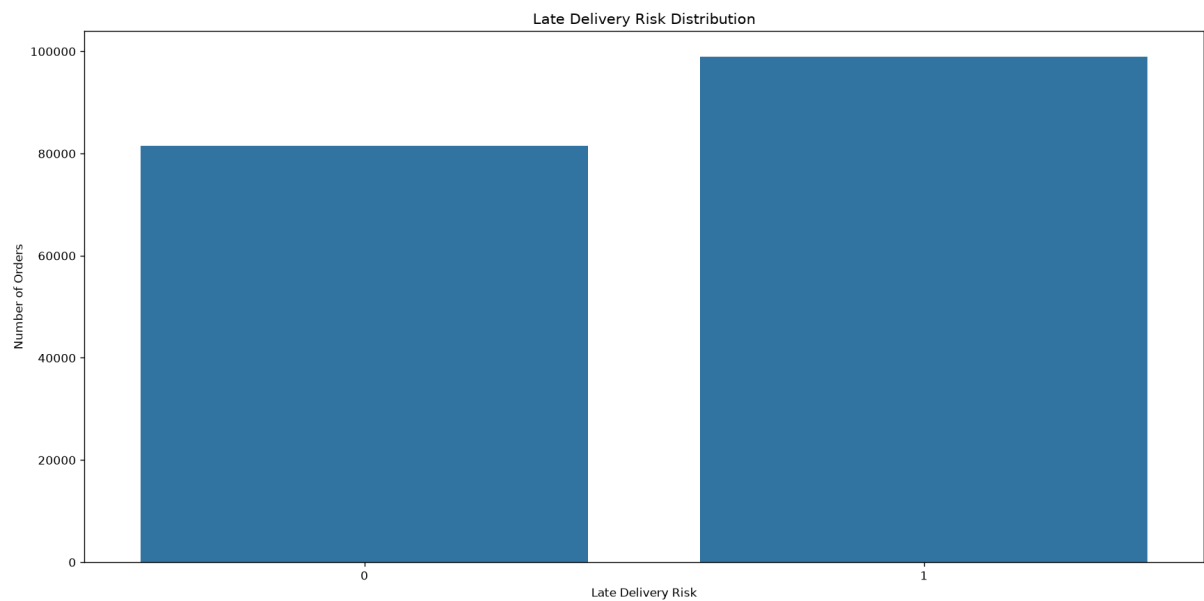


Figure 1. Distribution of late-delivery risk.

The late-risk class is slightly larger than the non-risk class, supporting binary classification without severe class imbalance.

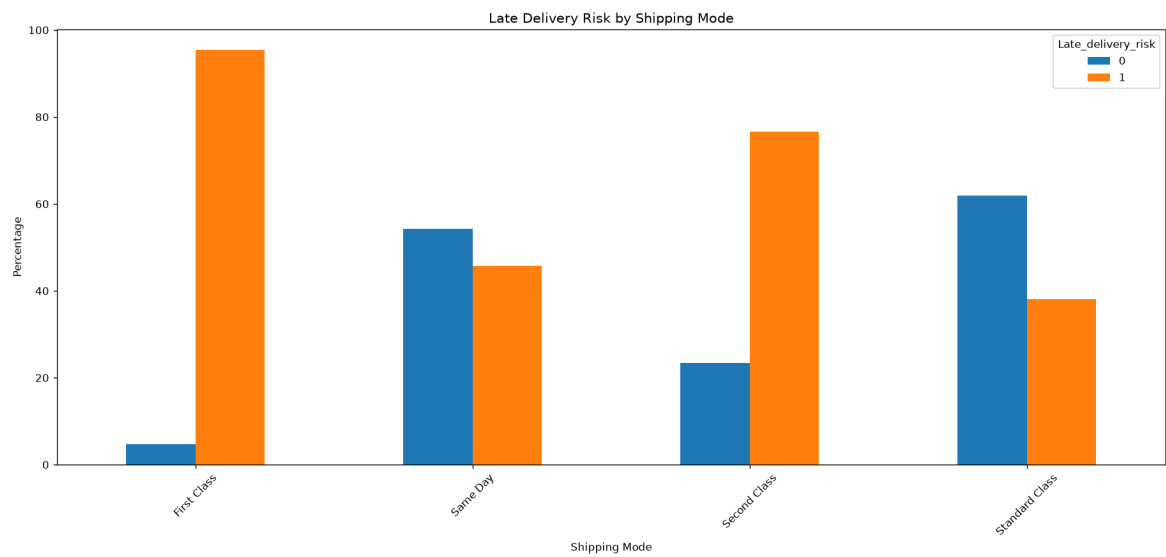


Figure 2. Late-delivery risk by shipping mode.

Risk varies substantially by shipping mode. First Class and Second Class show high proportions of the late-risk class in the plotted data, while Standard Class is lower.

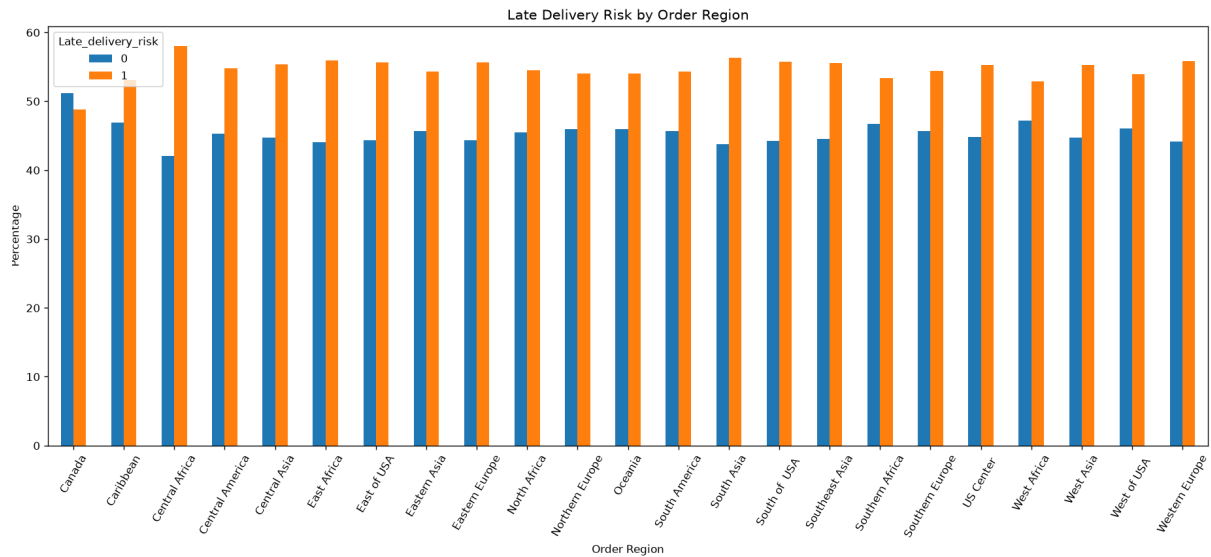


Figure 3. Late-delivery risk by order region.

The risk distribution differs across geographic regions, suggesting that geography can capture operational and network differences.



Figure 4. Late-delivery risk by customer segment.

Consumer, Corporate, and Home Office segments show broadly similar but not identical risk distributions.

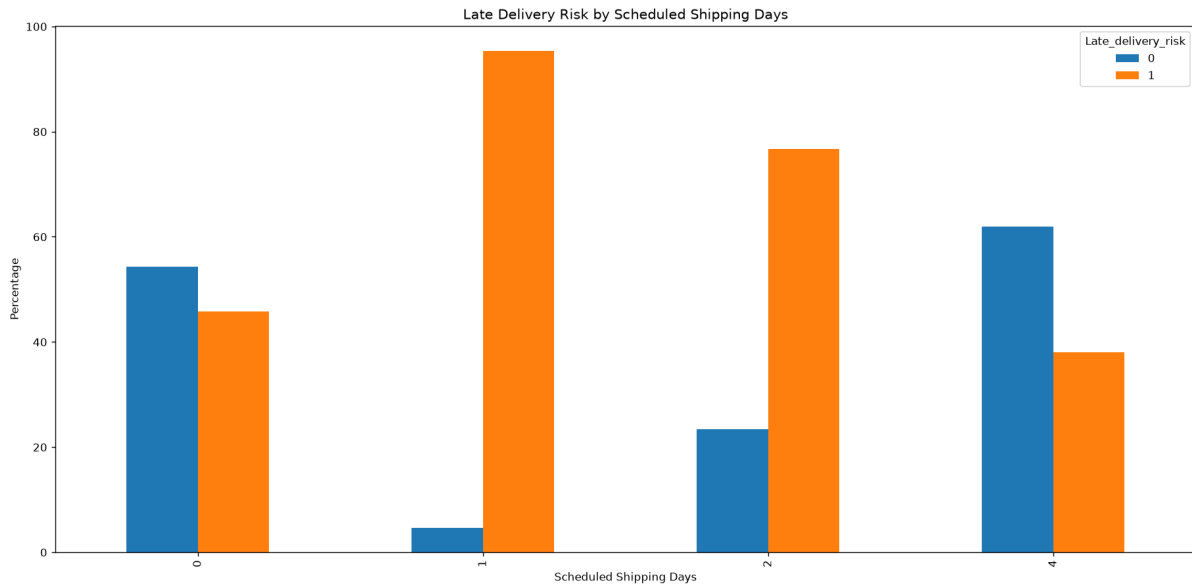


Figure 5. Late-delivery risk by scheduled shipping days.

Scheduled shipping time shows strong variation in risk proportions and is therefore retained as an important predictive variable.

6. Feature Engineering

Six engineered features were created to represent operational relationships:

Feature	Business interpretation
Shipping_Pressure	Order quantity divided by scheduled shipping days plus one.
High_Quantity_Flag	1 when order quantity exceeds the dataset median.
High_Discount_Flag	1 when discount rate exceeds 0.20.
Express_Shipping_Flag	1 when shipping mode contains 'express'.
Order_Complexity	Order quantity multiplied by one plus discount rate.
Profit_Margin	Order profit relative to absolute sales, with one added to avoid division by zero.

7. Methodology

Thirty predictors were used after feature selection and engineering. An 80:20 stratified split with random_state=42 produced 144,415 training records and 36,104 test records. Preprocessing was placed inside model pipelines to ensure consistent transformations during training and inference.

7.1 Models

Logistic Regression was used as a linear baseline. **Random Forest** provided a nonlinear ensemble benchmark. **XGBoost** was used as the main boosted-tree model; XGBoost is a scalable tree-boosting system designed for efficient learning on large datasets [1].

7.2 Evaluation

ROC-AUC was the primary model-ranking metric. Precision measures the proportion of predicted-risk cases that are truly positive; recall measures the proportion of actual positive cases detected; F1-score balances precision and recall; accuracy is reported as an overall descriptive metric.

8. Model Training and Results

Model	ROC-AUC
Logistic Regression	0.7552
Random Forest	0.7536
XGBoost	0.7601

XGBoost achieved the highest ROC-AUC of 0.7601 and was selected as the final model.

Metric	Result
ROC-AUC	0.7601
Precision	0.8449
Recall	0.5574
F1-score	0.6717
Accuracy	70.13%
Test observations	36,104

The model has high precision but lower recall. This means flagged orders are often truly risky, while a meaningful number of actual late-risk orders are missed at the default threshold. Operational threshold selection should therefore reflect the cost of missed late deliveries.

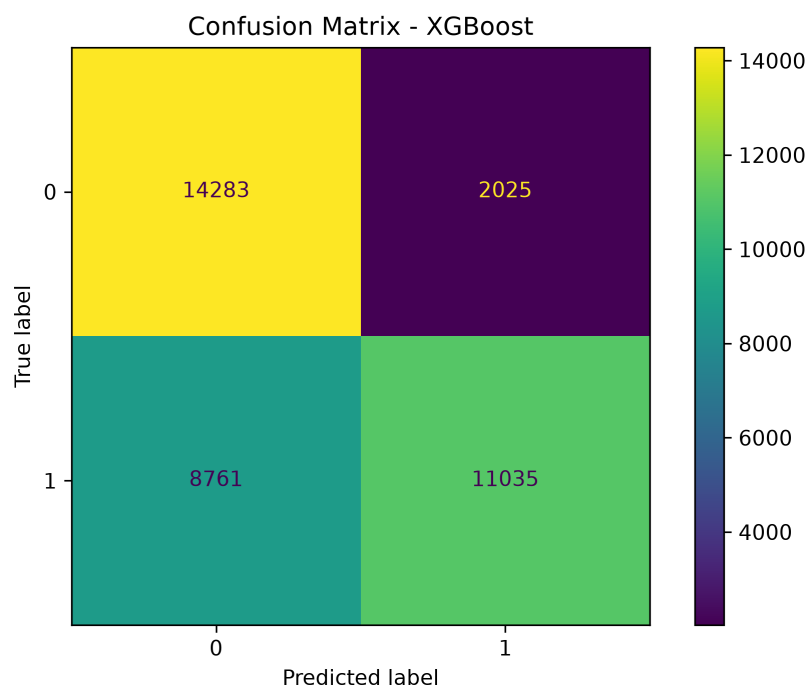


Figure 6. Confusion matrix for the final XGBoost model.

The confusion matrix contains TN=14,283, FP=2,025, FN=8,761, and TP=11,035. The total is 36,104 test observations. These values are consistent with the reported precision, recall, F1-score, and 70.13% accuracy.

9. Explainable AI Using SHAP

SHAP (SHapley Additive exPlanations) provides feature-attribution values that quantify how model inputs contribute to predictions. The SHAP framework was introduced as a unified approach to interpreting model predictions [2].

For this project, SHAP was calculated on a 1,000-observation sample after preprocessing. Because categorical variables are one-hot encoded, the displayed feature names are transformed indicators rather than the original high-level categorical fields.

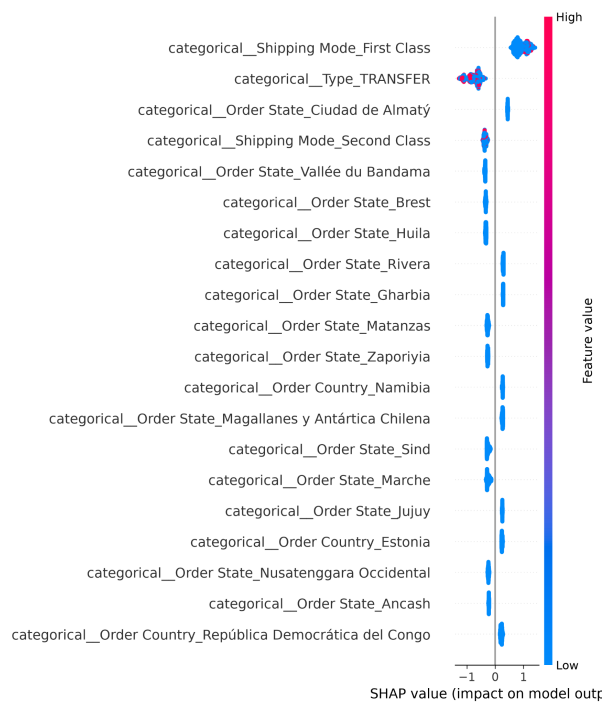


Figure 7. SHAP summary plot for the XGBoost model.

Shipping Mode and Type-related transformed indicators appear among the influential features in the displayed sample. These should be interpreted as category-specific model contributions rather than causal effects. Grouping one-hot contributions back to the original business variables would make future explanations easier for stakeholders to interpret.

10. Risk Classification and Streamlit Dashboard

The dashboard converts predicted probability into illustrative categories: Low (<0.33), Medium (0.33 to <0.66), and High (≥ 0.66). These thresholds are decision-support defaults, not statistically optimized operating thresholds.

The Streamlit application accepts selected operational and commercial inputs, applies the same feature-engineering logic used during model development, and returns predicted probability and risk category.

11. Business Implications and Recommendations

- Prioritize high-risk orders through an exception queue.
- Review shipping-mode performance and service-level behavior.
- Use scheduled shipping-time patterns to identify vulnerable service windows.
- Monitor geographic regions with elevated risk.
- Trigger proactive customer communication for high-risk orders.
- Use SHAP explanations to support targeted analyst review.
- Optimize the operating threshold using explicit business costs.

12. Public-Sector and Government Relevance

A predictive logistics-risk framework can support public-sector supply-chain monitoring for time-sensitive goods, infrastructure materials, and other critical distribution activities. Risk scoring can support early-warning dashboards, regional monitoring, service-level oversight, and prioritization.

For public-sector use, the model should remain a decision-support system with human review, auditability, data-quality controls, threshold governance, and periodic validation.

13. Limitations

- The model uses historical observational data; predictive associations are not causal claims.
- ROC-AUC of 0.7601 indicates useful but imperfect discrimination.
- Recall of 0.5574 means many actual late-risk cases are missed at the default threshold.
- The SHAP chart uses transformed one-hot encoded features.
- Dashboard thresholds are illustrative and require business calibration.
- External validation on a different period, carrier, geography, or network was not performed.

14. Future Scope

- Optimize decision thresholds using false-positive and false-negative business costs.
- Use time-based and rolling-window validation.
- Add carrier, route, weather, holiday, congestion, warehouse, and real-time tracking variables where available.
- Group SHAP contributions into original business variables.
- Evaluate probability calibration.
- Compare additional boosting and cost-sensitive algorithms.
- Monitor data drift, concept drift, and operational outcomes after deployment.

15. Conclusion

This study developed an end-to-end machine-learning framework for late-delivery risk prediction using 180,519 logistics records. The project integrates EDA, leakage-aware feature selection, business feature engineering, model comparison, evaluation, explainability, and a Streamlit decision-support interface.

XGBoost achieved the best ROC-AUC of 0.7601. Its test precision of 0.8449, recall of 0.5574, F1-score of 0.6717, and accuracy of 70.13% demonstrate useful predictive capability while also showing why threshold selection matters for operational deployment.

The framework provides a practical foundation for shifting logistics management from reactive reporting toward proactive risk identification. Further validation, richer real-time data, calibration, and monitoring would be needed for production deployment.

16. References

- [1] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 785–794. <https://doi.org/10.1145/2939672.2939785>

- [2] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* 30.
- [3] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [4] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232.

17. Reproducibility

The project contains reusable feature-engineering code, model training and evaluation scripts, the trained model artifact, SHAP analysis, Streamlit application, requirements file, README, and report figures. The reported model evaluation uses a fixed random seed and stratified 80:20 split.

Figure 1 — Late Delivery Risk Distribution

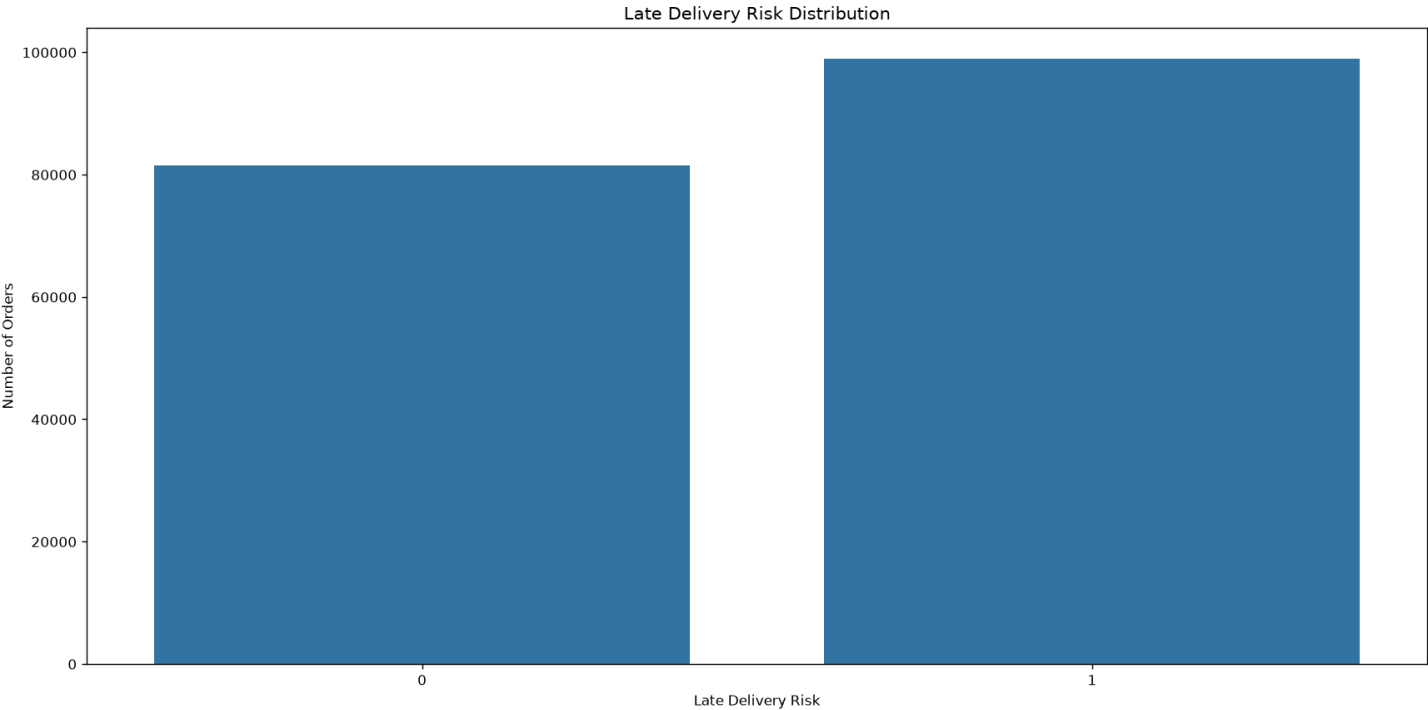


Figure 2 — Late Delivery Risk by Shipping Mode

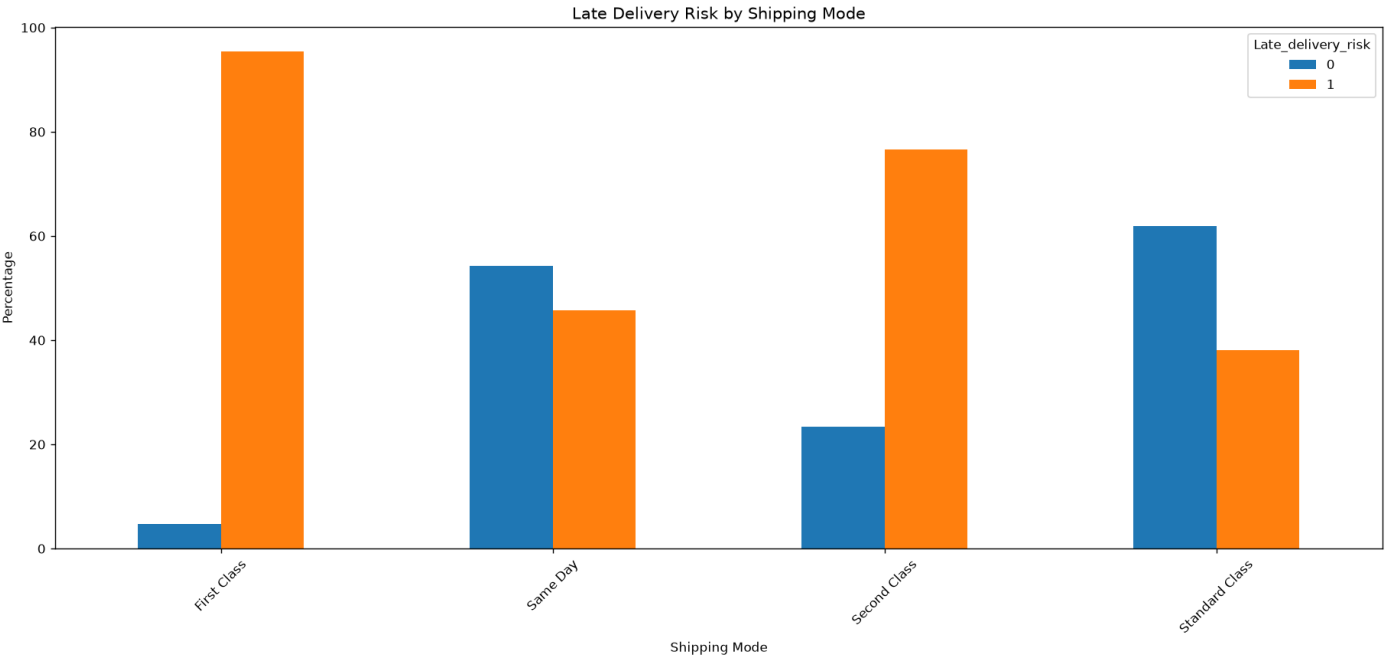


Figure 3 — Late Delivery Risk by Order Region

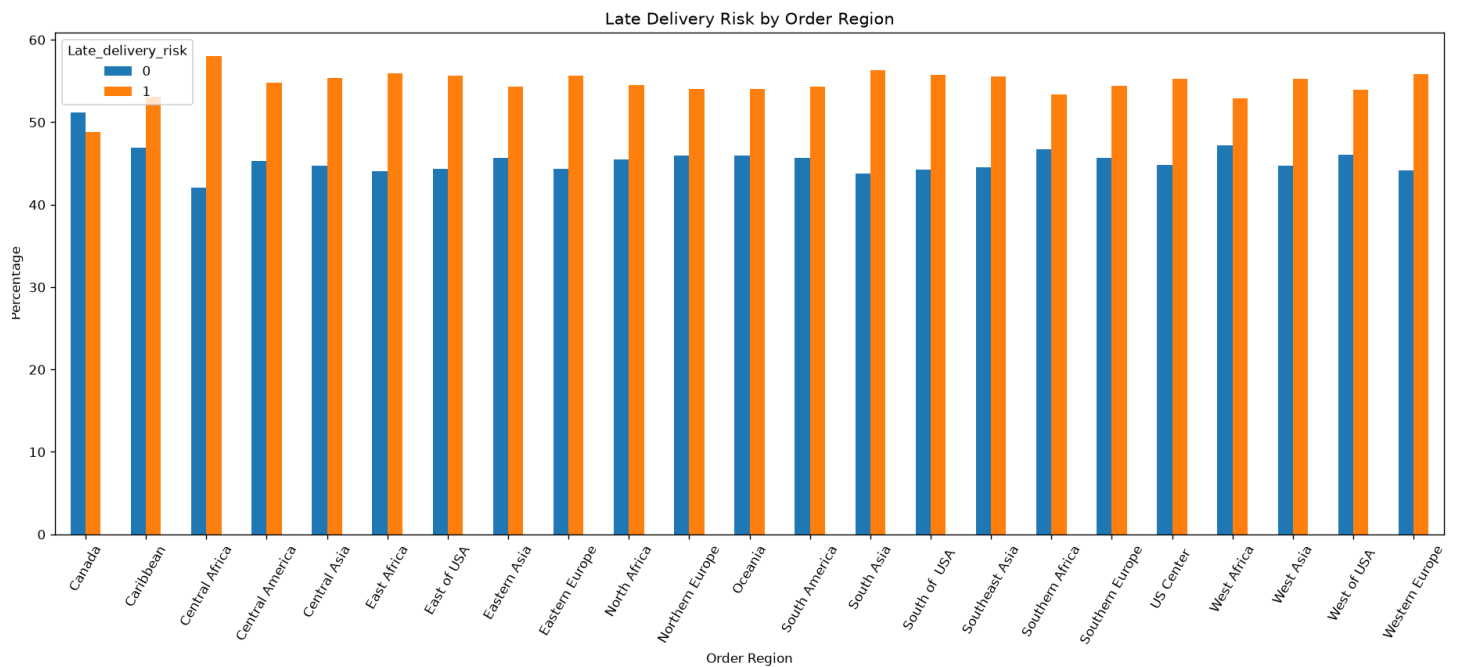


Figure 4 — Late Delivery Risk by Customer Segment

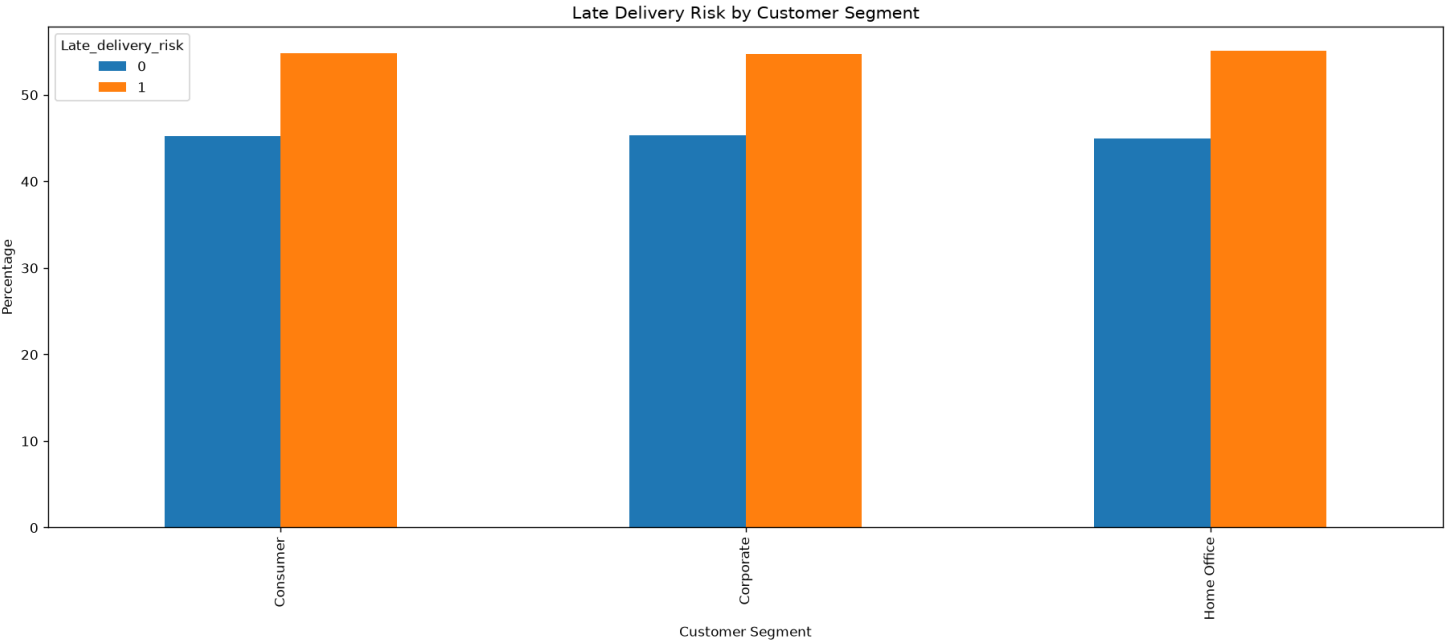


Figure 5 — Late Delivery Risk by Scheduled Shipping Days

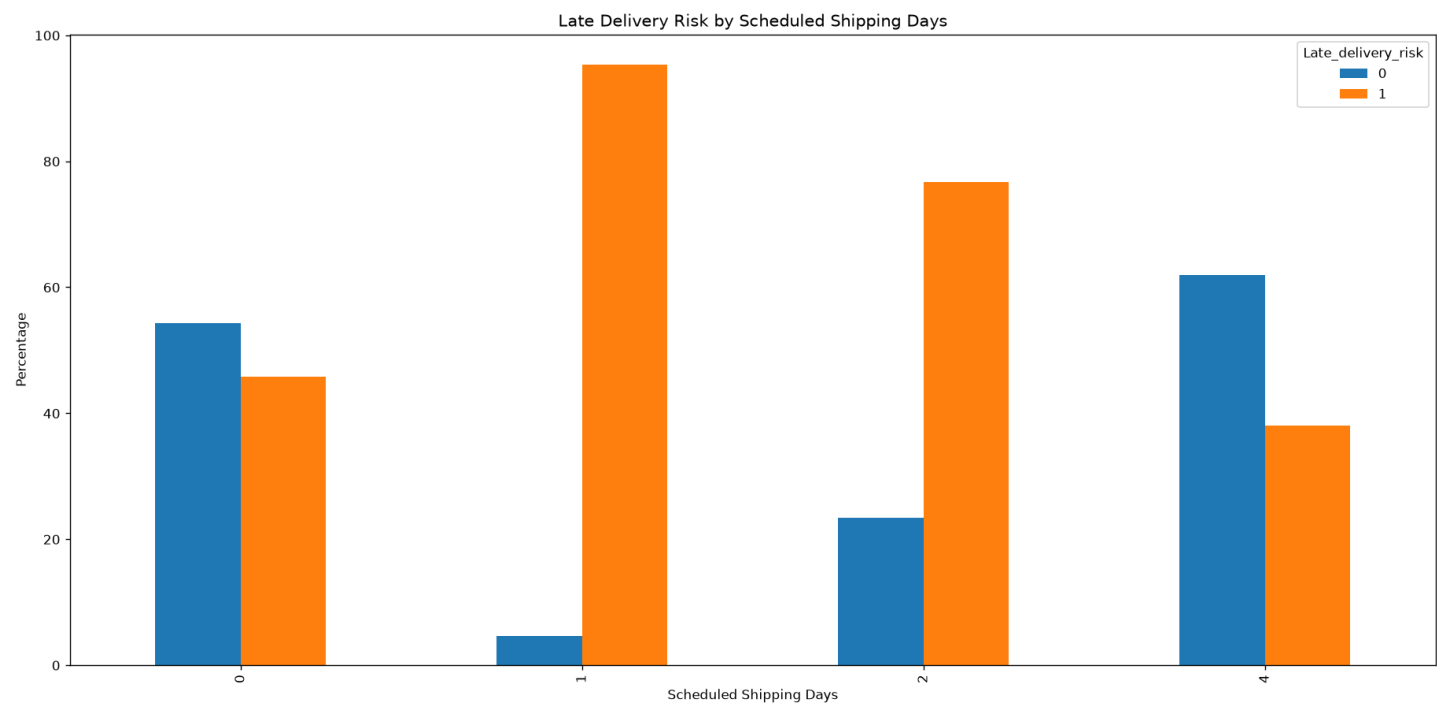


Figure 6 — Confusion Matrix — XGBoost

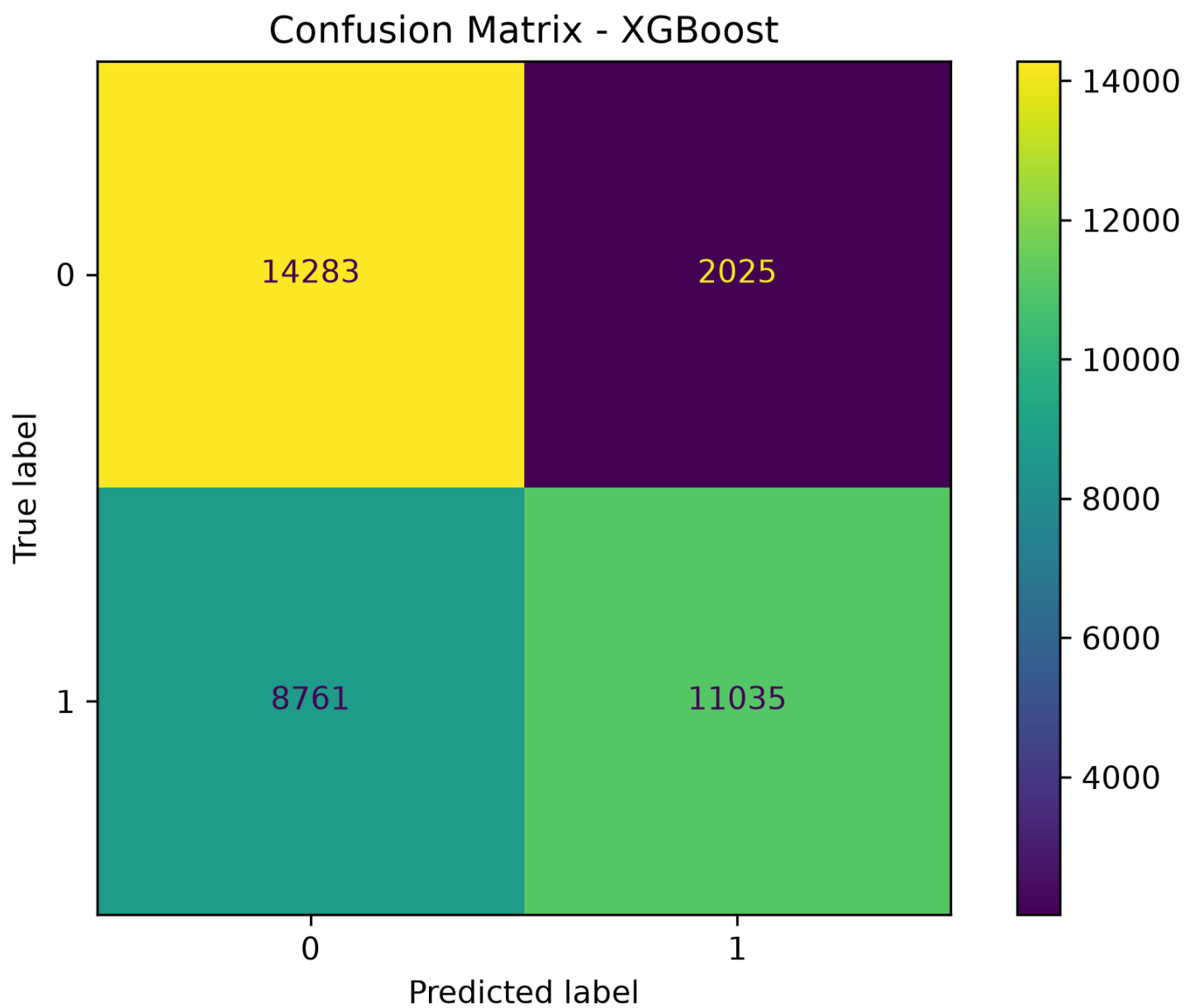


Figure 7 — SHAP Summary Plot — XGBoost

